

Algebra II with Trigonometry

Chapter 11.1

Mental Math

Saved tutoring response with MathJax rendering. Original PDF file:
[4652951_alg-2trig-h-chp-11-1-note-docx.pdf](#)

Certainly! Here are **10 mental math questions** designed to target key concepts from the PDF, all answerable without pen and paper.

1. Concepts: Center of Circle

What do you call the fixed point from which every point on a circle is equidistant?

2. Circle Equation

If a circle is centered at the origin with radius 4, what is its equation?

3. Parabola Shape

Does the graph of $y = 2x^2 + 3x + 1$ open upwards or downwards?

4. Axis of Symmetry for Parabola

If a parabola's vertex is at the origin and the equation is $x^2 = 8y$, what axis is the parabola symmetric about?

5. Parabola Direction

For the equation $y^2 = -12x$, does the parabola open to the left or the right?

6. Focus Location

For a parabola with equation $x^2 = 4py$ and vertex at $(0,0)$, what is the y-coordinate of its focus?

7. Directrix Equation

If a standard vertical parabola has focus $(0,5)$, what is the equation of its directrix?

8. Recognizing Parabola Orientation

True or False: The parabola $y^2 = 16x$ has its axis parallel to the x-axis.

9. "Width" of Parabola

What is the length of the latus rectum (focal diameter) of the parabola $x^2 = 12y$?

10. Opening of Parabola

If $p < 0$ in the equation $y^2 = 4px$, in which direction does the parabola open?

Let me know if you need answers, explanations, or more challenging questions!